CSE 20 - Discrete Mathematics - Jones [FA23]

Discrete Math for CS

(updated 9/28. Subject to change)

Content Calendar (https://canvas.ucsd.edu/courses/48949/pages/content-calendar)

Annotated Lecture notes (https://canvas.ucsd.edu/courses/48949/pages/annotated-lecture-notes)

Discussion Slides (https://canvas.ucsd.edu/courses/48949/pages/discussion-slides?wrap=1)

Homeworks (https://canvas.ucsd.edu/courses/48949/pages/homeworks)

Office Hours Calendar (https://canvas.ucsd.edu/courses/48949/pages/ta-weekly-office-hours)

One-on-one Tutoring Sign Up (https://docs.google.com/spreadsheets/d/1L83psvNLqF8YI5-l96ieVrpf0JU6CmAFYT_7J-0iOBc/edit#gid=1814389710)

CSE 20 FALL 2023

Welcome to CSE20! If you ever wondered "What sort of mathematics do I need for computer science?", this course will provide some of the answers. In particular, you will have the opportunity to learn basic concepts about algorithms, computer arithmetic, number systems, Boolean algebras, logic, proofs, program correctness, modular arithmetic, linear and partial orders, recurrences, and induction, amongst other things.

In this class, you will use careful mathematical modeling, problem solving, and clear and precise communication to explore key questions in Computer Science. (1) How do we decide (and prove) what's true? (2) How do we use mathematics to give multiple representations of data and computation?

Prerequisites

There are no strict prerequisites but students should have a basic understanding of elementary computer programming, mathematics (algebra, geometry, trigonometry and calculus.)

Course Logistics:

Time and Location

Date	Day	Time	
Lecture	Tu/Th	9:30-10:50 am	MOS 114

Discussion Sections	Fridays	10:00-10:50 12:00-12:50	CENTER 109 CENTER 214
Final Exam	Thursday, December 14	Scheduled time: 8:00am-10:59am	WLH 2001

Lectures and review quizzes:

- The lectures will be taught *in-person* in the lecture hall on the schedule of classes (unless otherwise specified.)
- The lecture slides/notes will be posted before class on the content calendar link.
- You may interrupt lecture to ask questions, make comments or express doubts.
- All lectures will be recorded and posted for students to watch when they want.
- There will be a short review quiz week, open until the Sunday of that week, You will have unlimited attempts on the review quizzes.

Instructor and Course Staff

Name	Role	email	OH time	OH location
Miles Jones	Instructor	mej016@eng.ucsd.edu	Tuesdays 2:30 -3:30 Mondays 2:30-3:30	CSE 4208 https://ucsd.zoom.us/j/99078066021 (https://u
Sam Chen	TA	sac003@ucsd.edu	Sundays (zoom): 4:00-5:00pm Wednesdays (in person): 10:00-11:00am **10/11 only: 9am-10am**	HDSI 432 **10/11 only: HDSI, Room 432** Zoom: https://ucsd.zoom.us/j/8377230947 (htt q=https://ucsd.zoom.us/j/8377230947&sa=D&source=
Robert Jiang	TA	rjiang@ucsd.edu		B260A

			Mondays: 12pm-2pm	
Natalie Norstad	Tutor	nnorstad@ucsd.edu	Mondays: 11am-12:30pm	B260A
Matthew Sun	Tutor	mrsun@ucsd.edu	Fridays: 1pm-2:30pm	B260A
Sean King	Tutor	sek032@ucsd.edu	Mondays: 11am-12:30pm	B260A
Henry Lin	Tutor	hhlin@ucsd.edu	Wednesdays: 12:00pm - 1:30pm	B270A
Oleg Bychenkov	Tutor	obychenkov@ucsd.edu	Tuesdays: 4:30pm - 6:30pm Finals week: Wednesday (Zoom): 7:00pm - 8:45pm	B260A **Zoom** (https://ucsd.zoom.us/j/4855594631?pwd=N
Nirmal Agnihotri	TA	nagnihot@ucsd.edu	Tuesdays: 5 -6pm Wednesdays: 4:30 - 6:30pm	Tuesdays: **https://ucsd.zoom.us/j/5759176413** Wednesdays: B240a

Resources

Link piazza: **https://piazza.com/ucsd/fall2023/cse20** (https://piazza.com/ucsd/fall2023/cse20)

Gradescope link: **https://www.gradescope.com/courses/629574** (https://www.gradescope.com/courses/629574)

(https://www.gradescope.com/courses/382177) Gradescope code: XXGW56

We encourage you to work on homework in groups of up to four CSE 20 classmates. To find group members: reach out to people sitting around you in class, in discussion section, or during office hours. Working within the campus safety guidelines, you may choose to meet with your group mates in person (find ideas for **where to study on campus (https://students.ucsd.edu/campus-services/technology/campus-study-locations.html)**), or online. If you're working with one another remotely, we highly recommend meeting synchronously so that you can work through the homework problems *together*.

Textbook and supplies

Previous versions of the course used reference textbooks, including *Discrete Mathematics and its Applications* by Kenneth Rosen, which has **online self-assessments and extra examples (https://highered.mheducation.com/sites/0073383090/student_view0/self_assessments.html)**. This quarter, I will integrate the material that has been curated by Mia Minnes and several students. The hope is that all of the material you need will be available to you through the website and notes.

Here is a link to her glossary of terms from when she taught the class previously:

https://discrete-math-for-cs.github.io/website/glossary.html (https://discrete-math-for-cs.github.io/website/glossary.html)

Here are some practice problem from the Rosen textbook that roughly correspond to the class:

https://sites.google.com/a/eng.ucsd.edu/cse-20-s2020/preclass-readings (https://sites.google.com/a/eng.ucsd.edu/cse-20-s2020/preclass-readings)

Typesetting (LaTeX) Resources

All submitted homework for this class must be typed. Diagrams may be hand-drawn and scanned and included in the typed document. You can use a word processing editor if you like (Microsoft Word, Open Office, Notepad, Vim, Google Docs, etc.) but you might find it useful to take this opportunity to learn LaTeX. LaTeX is a markup language used widely in computer science and mathematics. The homework assignments are typed using LaTeX and you can use the source files as templates for typesetting your solutions.

If you have never used LaTeX, we recommend cloud resources that don't require you to download and install LaTeX on your local machine. A good example is **Overleaf (https://www.overleaf.com/)**, which has lots of **documentation (https://www.overleaf.com/learn)**. Overleaf works similar to Google Docs in that all members can edit the file in parallel and changes are updated in real time. There is a way to directly invite group members to your document, but the free version of Overleaf only allows two people to work at the same time. To get around this, turn on link sharing: Click on “Share” in the top right, Click “Turn on link sharing”, Copy the displayed link and share it with your group members. To export your work, click on the “Download PDF” button on the right-hand side If you want to export the raw source files, click on the “Menu” button in the top-left, then click on “Source”.

This **open source LaTeX reference (http://en.wikibooks.org/wiki/LaTeX)** can be helpful when getting started, and you can use the .tex source of all the files we use in class as templates.

Alternatively, you can use Google Docs, which is available through your @ucsd.edu account. You can create documents and then share them with your group members with manual invites or a shareable link. Google Docs has a LaTex add-on that lets you type formulas in a math typesetting environment: search for "Auto-LaTeX Equations" if you want to try this option. You'll need to use the display environment (start and end with $) for all the portions you want rendered with LaTeX.

You will be deducted 1 point per handwritten question.

Grading and assignments

Grades in this class are designed to reflect your work and to document evidence of your learning this core material. By working together to explore the CSE 20 material with integrity, we can each help ensure a **fair** and **interesting** quarter of building the foundations needed for your continued development in Computer Science. Please reach out to me (mej016@eng.ucsd.edu) if you have extenuating circumstances that you think will impede your ability to participate in the planned CSE 20 activities; I'd like to work out a solution together.

The graded components for CSE 20 will be Review quizzes, Homework, tests, midterm, and Final exam. Your overall grade for CSE 20 will be computed using the weights

- **Review quizzes**: 5% of overall score.

  Why? To help you track and confirm your understanding of the concepts and examples we work in class and give you immediate feedback.
  How? Quizzes can be submitted as many times (with no penalty) as you like until the quiz deadline: the two quizzes each week are all due on Sunday. The lowest three quiz scores will not be included in your overall review quiz average. You can collaborate with other CSE 20 students on review quizzes and you can ask questions about them in public posts on Piazza and in office hours.
  - There will be a total of 10 review quizzes.
  - You will get the full 5% if you complete 8 or more.
  - For each number below 8, you will lose 1%.
  - 8+: 5%
  - 7: 4%
  - 6: 3%
  - 5: 2%
  - 4: 1%
  - 3-: 0%
- **Homework**: 31% of overall score.

  Why? To give you practice with the main concepts and techniques of the course, while getting to know

and learn from your classmates.
How? Weekly homework may be done individually or in groups of up to 4 students. There will be 8 homework assignments. You may switch HW partners for different HW assignments. The lowest HW score will not be included in your overall HW average. Please ensure your name(s) and PID(s) are clearly visible on the first page of your homework submission and then upload the PDF to **Gradescope (http://gradescope.com/)**. If working in a group, **submit only one submission per group**: one partner uploads the submission through their Gradescope account and then adds the other group member(s) to the Gradescope submission by selecting their name(s) in the "Add Group Members" dialog box. You will need to re-add your group member(s) every time you resubmit a new version of your assignment. You may only collaborate on HW with CSE 20 students in your group; if your group has questions about a HW problem, you may ask in office hours or post a private post (visible only to the Instructors) on Piazza.

Late policy: We will accept late homework for a -1 point penalty until 5:00pm on the day after the due date.

- **Tests:** 8% each.

  Why? To give you the opportunity to understand what you know and what you don't know.
  How? The tests will be administered on Canvas. The questions are randomly selected from a question bank. You will have 60 minutes to complete the test during the day of the test (24 hour window.) After the test results have been posted, you will have the option of doing a "re-test" where you can earn up to 1/3 the points you missed from the original test. You may openly ask for help for the "re-test" and you will have unlimited attempts throughout the weekend.
  The tests will be administered all day Friday and you can take it at any time during the day (12:01am to 11:59pm.) I understand that this may conflict with some of your schedules so if you have a conflict, then please let me know and we can do our best to accommodate you.
- **Midterm:** 18%

  Why? To give us the opportunity to assess your understanding of the first half of the content.
  How? The midterm will be administered during the scheduled classtime (80 minutes). The midterm will be administered in-person. (Please contact us if you are sick or have an emergency that prevents you from coming to class in person. We will do our best to work out a solution.) The midterm is open-notes (no phones.) It is forbidden to discuss the midterm while it is happening. There will not be a "re-midterm"
- **Final**: 30% of overall score.
  *Why? T*o review the entire quarter and solidify your understanding of CSE 20's foundational material.
  How? Cumulative test on class content during exam week. The exam must reflect your own independent effort. The exam may have a combination of True/False, multiple choice, select-all-that-apply, and free-response questions.

Grading option 1:

Review quizzes:	**5%**
HW 0:	**1%**
HW1-8:	**30% (best 7 of 8)**
Test 1:	**8%**
Test 2:	**8%**
Midterm:	**18%**
Final:	**30%**

Grading option 2:

Review quizzes:	**5%**
HW1-8:	**30% (best 7 of 8)**
Test 1:	**10%**
Test 2:	**10%**
Midterm:	**0%**
Final:	**45%**

Grading option 3:

Review quizzes:	**5%**
HW1-8:	**30% (best 7 of 8)**
Best of Test 1 and 2:	**10%**
Midterm:	**15%**
Final:	**40%**

Grading option 4:

Review quizzes:	**5%**
HW 0:	**1%**
HW1-8:	**30% (best 7 of 8)**
Test 1:	**10%**
Test 2:	**10%**
Midterm:	**0%**
Final:	**44%**

Grading option 5:

Review quizzes:	**5%**
HW 0:	**1%**
HW1-8:	**30% (best 7 of 8)**
Best of Test 1 and 2:	**10%**
Midterm:	**15%**
Final:	**39%**

Grade Scale: Your final grade will be based on the following scale. (You will earn the grade in the table based on your numerical score or higher.)

A+	A	A-	B+	B	B-	C+	C	C-	D	F

98	93	90	86	82	78	74	70	64	60	below 60

Academic Integrity

The UC San Diego Academic Integrity pledge is **here (https://academicintegrity.ucsd.edu/forms/form-pledge.html)** . Academic integrity violations will be taken seriously and reported to the campus-wide Academic Integrity Office. Key facts about academic integrity related to CSE 20:

- Use only resources explicitly allowed for each assignment. Resources not affiliated with this quarter's version of the class may use inconsistent notation or definitions. If you need help, please reach out to the instructor, TAs, and tutors.
- Do not share written solutions or partial solutions for homework with other students in the class who are not in your group. Doing so would dilute their learning experience and detract from their success in the class. You are free to share hints and basic techniques with others outside of your group.
- Before and during taking any individual assessment, do not attempt to obtain information about the contents of the exam from students who have already taken it or from any nonauthorized source.
- You may not ask for help from anyone while taking individual assessments since they are intended to reflect *your own* mastery of the material. In particular, you may not collaborate on exam questions with other students in the class and you may not post any portion of the exam on forums where others may assist you.
- After taking an exam or quiz, do not discuss its contents with anyone in the class who has not yet taken it. Do not post information about it or share information about it with others who haven't taken it.

UCSD has fantastic resources to support your learning, with integrity. Of course, the instructional team for CSE 20 is here to help you navigate the course content. The **Jacobs School of Engineering IDEA Center (http://jacobsschool.ucsd.edu/idea/)** organizes group study sessions and can connect you with student organizations. The **Teaching and Learning Commons (https://commons.ucsd.edu/)** continues to offer their full suite of student success programs.

Diversity and Inclusion

We are committed to fostering a learning environment for this course that supports a diversity of thoughts, perspectives and experiences, and respects your identities (including race, ethnicity, heritage, gender, sex, class, sexuality, religion, ability, age, educational background, etc.). Our goal is to create a diverse and inclusive learning environment where all students feel comfortable and can thrive.

Our instructional staff will make a concerted effort to be welcoming and inclusive to the wide diversity of students in this course. If there is a way we can make you feel more included please let one of the course

staff know, via email/discussion board or during/after lecture/office hours. Our learning about diverse perspectives and identities is an ongoing process, and we welcome your perspectives and input.

We also expect that you, as a student in this course, will honor and respect your classmates, abiding by the UCSD Principles of Community (https://ucsd.edu/about/principles.html). Please understand that others' backgrounds, perspectives and experiences may be different than your own, and help us to build an environment where everyone is respected and feels comfortable.

If you experience any sort of harassment or discrimination, please contact the instructor as soon as possible. If you prefer to speak with someone outside of the course, please contact the Office of Prevention of Harassment and Discrimination: https://ophd.ucsd.edu/.

Students with Disabilities

We aim to create an environment in which all students can succeed in this course. If you have a disability, please contact the Office for Students with Disability (OSD), which is located in University Center 202 behind Center Hall, to discuss appropriate accommodations right away. We will work to provide you with the accommodations you need, but you must first provide a current Authorization for Accommodation (AFA) letter issued by the OSD. You are required to present their AFA letters to Faculty (please make arrangements to contact me privately) and to the OSD Liaison in the department in advance so that accommodations may be arranged.

Basic Needs/Food Insecurities

If you are experiencing any basic needs insecurities (food, housing, financial resources), there are resources available on campus to help, including The Hub and the Triton Food Pantry. Please visit http://thehub.ucsd.edu/ for more information.

Course Summary:

Date	Details	Due